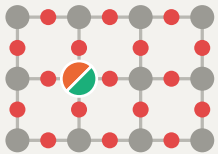
Pr and Rb lower the O-vacancy cost; only Rb makes it spontaneous

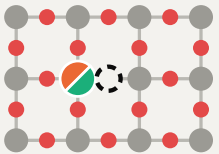
Anatase TiO_2, PBE+U, neutral V_O, O-rich limit $\mu_O = E(O_2)/2$

❶ Dope



Ti → Pr^{3+} or Rb^+;
1 per cell,
48 atoms

❷ Remove O



Neutral V_O releases ●● 2 e^-

❸ Protocol

U(Ti-3d) 3.5 eV
Γ-only, 40 Ry

✓ k-mesh, cut-off, D3: Pr–Rb Δ shifts ≤ 0.04 eV

! smearing: Pr cells polarise, Pr +0.99 eV

❹ Bookkeeping

○ h^+ ● e^- (formal)

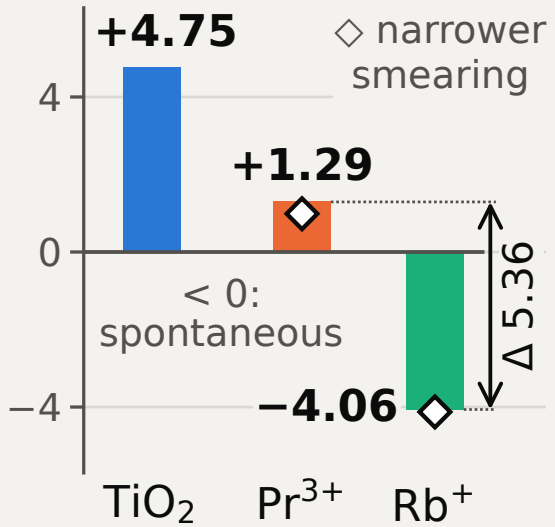
● Pr^{3+}

○ + ●● → ●

1 e^- left: on Ti-3d

● Rb^+

○○○ + ●● → ○

1 h^+ left: on O-2p

❺ Result: E_f (eV)